OSTEOARTHRITIS INITIATIVE / SECONDARY LONGITUDINAL ANALYSIS

Structure-Pain Discordance and 48-Month Knee Pain Prediction Error

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8,232
knees with 48-month outcomes

1.787
held-out MAE (WOMAC points)

1.582
retained MAE after 10% deferral

MAIN INTERPRETATION

The unadjusted association did not persist after baseline-pain adjustment. Symptom burden is the stronger candidate explanation in these data.

01 / QUESTION & DESIGN

Does mismatch identify harder-to-predict knees?

Knee X-ray severity and reported pain often disagree. We tested whether this baseline mismatch tracked absolute error when predicting 48-month WOMAC pain.

COHORT

8,920 baseline knees

4,490 participants; baseline KL grade and pain available; baseline replacements excluded; Project 15 radiographic readings only.

8,232 knees / 4,146 participants had observed 48-month WOMAC pain. Outcomes were not imputed.

LEAKAGE-PROTECTED ANALYSIS

1. Estimate expected baseline pain by KL grade inside training folds.

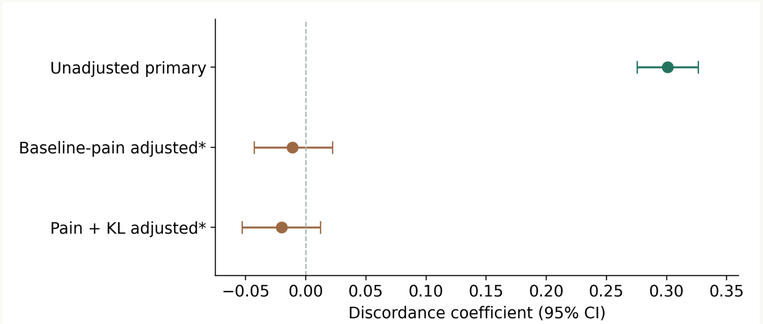
2. Define absolute discordance as the absolute gap between observed and expected baseline pain.

3. Fit elastic net using participant-grouped nested five-fold cross-validation; keep both knees of a person together.

4. Regress held-out absolute error on discordance using participant-clustered inference.

02 / ASSOCIATION

Adjustment changes the conclusion.



Frozen unadjusted primary

Coefficient 0.301; 95% CI 0.275 to 0.326. Positive association with held-out absolute error.

Post-freeze pain adjustment*

Coefficient -0.011; 95% CI -0.043 to 0.022; p = 0.52. No independent association detected.

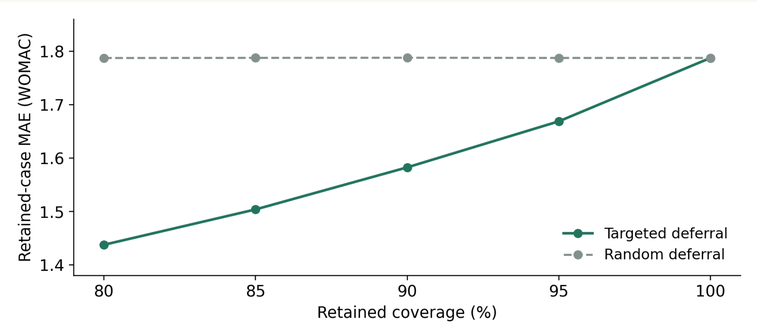
Why this matters

Discordance correlated with baseline pain (r = 0.751). Pain alone explained more error variance than discordance alone (R^2 0.202 vs 0.111).

*Adjusted analyses are post-freeze sensitivities reported in the manuscript, not prespecified. They do not replace the frozen primary analysis.

03 / SELECTIVE PREDICTION

Lower error, fewer retained cases.



At the frozen 10% uncertainty-deferral policy, 7,409 knees were retained. MAE fell from 1.787 to 1.582: an 11.47% relative reduction within that subset.

The uncertainty model included discordance. Its performance is not independent confirmation of a discordance effect. Subgroup deferral rates varied; report who is set aside.

LIMITS & NEXT STEP

One cohort; internal validation only. Baseline-pain adjustment was post hoc. Follow-up was missing for 7.7% of baseline knees. Quartile error means were not monotonic. No causal or clinical-utility claim is established. External validation is needed.

Research companion & reproducibility

ksricharan7.github.io/oai-model-reliability/

Sources: supplied manuscript, Table 3 and post-freeze sensitivity section; frozen aggregate model and risk-coverage outputs. Original results preserved. Poster prepared October 2026; no publication acceptance or prior presentation claimed. Research use only.